

NFL Franchise Valuation

By Andrew Wolfe

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NFL Franchise Valuation Context

- **Context:** This project investigates and predicts the value of NFL franchises. Factors included in this project are financial performance, market characteristics, on-field performance metrics, and brand equity. The goal of this project is to identify mispriced teams / assets from the viewpoint of a private equity acquisition in the future.
- **Dataset:** There were various datasets used in this project, including Forbes, nflverse, and supplementary hardcoded data from online sources. Features were created to reflect private equity values including the impact of the brand, economic value, and recent growth trajectory of NFL Franchises.
- **Machine Learning Models Used:** Random Forest, Gradient Boosting, and Ridge Regression models were tested. The models above all used a train – test split and a Leave-One-Out Cross (LOOC) Evaluation method. The purpose of the LOOC is to provide an unbiased and accurate estimate on unseen data where there is a low sample size. In this case we have a small sample size of $n = 32$ teams. R^2 (coefficient of determination) and the Mean Absolute Error were used as evaluation metrics.
- **Future Implications:**
 - Would incorporating debt load and new stadium pipeline data meaningfully improve mispricing signals?
 - Can this model be extended to MLB, NBA, and NHL Franchises to enable valuation benchmarking?
 - How would adding media right deals expirations affect inflection points in franchise values in the future?

Input and Target Variables

Revenue_mm	Total annual franchise revenue expressed in millions of dollars. This includes ticket sales, merchandise, sponsorships, and TV deals.
Op_income_mm	The operating income. This is equal to revenue – operating expenses. This was measured in millions of dollars and is very close approximation to EBITDA.
rev_cagr_5yr	The 5-year compound annual growth rate of revenue from 2019 to 2025. This captures the franchise growth trajectory.
social_following_mm	The combined total of Instagram and twitter followers in millions. This is very relevant for sponsorship and merchandise monetization upside.
Tv_market_rank_inv	The Nielsen DMA television market rankings inverse. Inverse was used so that higher values could indicate better markets, creating a linear relationship with total valuation.
Win_pct_5yr	Win percentage during the regular season averaged across the 2019-2025 seasons.
Stadium_owned	Binary indicator representing who truly owns the stadium lease (1 = under team or owner control).
Avg_attendance	Average home game attendance across the 2019-2025 seasons.
<u>Valuation_nm</u>	Forbes-report franchise equity value in millions of dollars. This is the target variable.

Evaluation Metrics and Results

R-Squared	The correlation coefficient. Explains the amount of variation used in the model. It ranges from 0 to 1, where higher numbers indicate better model evaluations. The R-Squared measures how much of a model's dispersion is explained.
MAE	Mean Absolute Error. The average number of dollars the model's predicted valuation missed the actual valuation expressed by Forbes (in millions).

Model	MAE	R^2
Ridge	\$800	0.672
<u>Random Forest</u>	\$641	0.699
Gradient Boosting	\$749	0.655

NFL Franchise Evaluation - Conclusion

Summary: A machine learning model and pipeline was created to evaluate and predict the true value of NFL Franchises. About eight features involving on-field results, financial performance, brand equity, social media popularity, and market quality were used. Data was from Forbes, nflverse, Wikipedia, and other supplementary websites. Random Forest, Ridge Regression, and Gradient Boosting Models were used in a 70-30 train-test split. The purpose of these models was to investigate and flag NFL Franchises valued above or below their true value.

Key Takeaways:

- Revenue and Operating Income explained over 65% of the model's predictive power.
- The Chargers and Browns were identified as the most undervalued franchises – their predicted valuations exceeded current market pricing.
- Stadium ownership ranked the lowest in terms of feature importance – indicating that building economics are not fully priced in the valuation market. This represents a potential inefficiency for private equity valuations in the future.

Next Steps:

- How would implementing a Leave-One-Out Cross Validation Model (running 32 separate models trained each time on 31 teams and predicating the one held out → rotating until every franchise is held out once) change the results of the project?
- As new franchises are added to leagues, how will this impact the model? At what correlation coefficient and MAE value does the model become reliable to place a bid in a live deal process?